

VINAYAK KOLI

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EDUCATION

Indraprastha Institute of Information Technology Delhi (IIIT-D)

2023 – 2027

Bachelor of Technology in Computer Science and Social Sciences

Delhi, India

Coursework: Database Management Systems, Object-Oriented Programming, Advanced Programming, Operating Systems, Data Structures and Algorithms, Algorithm Design, Collaborative Filtering

SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL

Frameworks and Libraries: React Native, FastAPI, PyTorch, LightGBM, LangChain, HuggingFace Transformers, Gradio, Electron

Tools and Platforms: Git, Appwrite, Expo, Optuna, Chrome Manifest V3, Vite, Zustand, Postman, VS Code, Android Studio

EXPERIENCE

AI Model Evaluation Research

Jan 2026 – Present

Undergraduate Researcher under Prof. Suman Roy / IIIT Delhi

- Designed and implemented a custom evaluation system for the DeepMind Code Contests dataset, enabling automated execution, correctness checking, failure detection, and structured result logging at scale.
- Benchmarked Small Language Models (SLMs) on DeepMind Code Contests and HumanEval datasets across 165+ problems, analysing pass@k, execution accuracy, and multi-step reasoning failure modes.
- Contributing to ongoing research on AI automation using SLMs for multi-step code generation and agentic reasoning workflows.

AI-Assisted Clinical Note Generation (Ongoing)

Apr 2026 – Present

Undergraduate Intern under Prof. Suman Roy / IIIT Delhi

- Building a system where multiple AI agents each playing a specialist role collaborate to convert doctor-patient conversations into structured clinical notes, debating their conclusions to reach a better diagnosis.
- Incorporating patient records and medical images alongside conversation text so the AI has fuller context; using techniques to reduce confident-sounding but incorrect outputs.
- Measuring how accurate and clinically useful the generated notes are; studying how AI agents can develop shared biases when they agree too quickly without sufficient independent reasoning.

PROJECTS

Hybrid Movie Recommendation System | Python, PyTorch, LightGBM, Gradio | [GitHub](#) | [Live Demo](#)

- Built a movie recommendation system that combines seven different AI models and merges their suggestions, so no single model's blind spots affect the final result, trained on over 1 million user ratings.
- Used a large language model to understand *why* a user might like or dislike something, then blended those insights with traditional recommendation signals for more personalised results.
- The system correctly recommended a movie the user actually wanted 80% of the time, with near-perfect personalisation scores and strong coverage of the full movie catalog.

ContextChat Desktop | Electron, React, TypeScript, Vite | [GitHub](#)

- Built a desktop AI assistant that lives in the system tray. Select any text or take a screenshot anywhere on screen and instantly chat with an AI about it, without switching apps or copying anything manually.
- Supports multiple AI providers (OpenAI, Anthropic, and locally-run Ollama models) with a built-in dashboard to manage API keys, check connection status, download models, and view conversation history.
- Added offline voice input using a locally-run speech recognition model. Press a shortcut, speak, and the transcript is sent directly to the chat panel, with no audio ever leaving the device.

Vocal for Local (VoLo) | React Native, FastAPI, Python | [GitHub](#) | [LinkedIn](#)

- Built a mobile app (iOS and Android) that helps users discover local businesses nearby, with Google sign-in, location-based search, and a clean 14-screen experience built from scratch.
- Added smart search that finds businesses by name, category, or location with server-side deduplication, and calculated real walking distances without relying on any third-party mapping library.
- Improved app response speed by about 35% by caching frequently-requested data on the server and making the UI load content progressively instead of all at once.

ACHIEVEMENTS AND ACTIVITIES

- **3rd Place** – DevHaven 2.0 Hackathon.
- Captain of Table Tennis Team for Karwaan 2024; Member of Sports Council Operations Team.
- Led a 10+ member team delivering a cancer awareness workshop impacting 100+ individuals at Tihar Jail.